5. **Photosynthesis** – Example: *Opuntia, Ruscus, Euphorbia.*
6. **Protection** – Example: *Citrus, Bougainvillea, Acacia.*
7. **Support** - Example: *Passiflora, Vitis, Cissus quadrangularis.*

3.6.1 Buds

Buds are the growing points surrounded by protective scale leaves. The bud primordium matures into bud. They have compressed axis in which the internodes are not elongated and the young leaves are closed and crowded. When these buds develop, the internodes elongate and the leaves spread out. Buds have architecture identical to the original shoot and develop into lateral branches or may terminate by developing into a flower or inflorescence. Based on origin, buds are classified into (a) Terminal or Apical bud (b) Lateral or Axillary or Axil bud. Based on function buds are classified into (a) Vegetative bud (b) Floral or Reproductive bud.

1. **Terminal bud or apical bud:** These buds are present at the apex of the main stem and at the tips of the branches.
2. **Lateral bud or Axillary bud:** These buds occur in the axil of the leaves and develop into a branch or flower.
3. **Extra axillary bud :** These buds are formed at nodes but outside the axil of the leaf as in *Solanum americanum.*
4. **Accessory Bud:** An extra bud on either side (collateral bud) or above (superposed bud or serial bud) the axillary bud. Example: *Citrus* and *Duranta*.
5. **Adventitious buds:** Buds arising at any part other than stem are known as **adventitious buds. Radical buds** are those that arises from the lateral roots which grow into plantlets. Example: *Millingtonia, Bergera koenigii (Murraya koenigii), Coffea arabica* and *Aegle marmelos*. **Foliar buds** are those that grow on leaves from veins or from margins of the leaves. Example: *Begonia* (Elephant ear plant) and *Bryophyllum* (Sprout leaf plant). **Cauline buds** arise directly from the stem either from cut, pruned ends or from branches. Adventitious buds function as propagules which are produced on the stem as tuberous structures. Example: *Dioscorea, Agave*.
6. **Bulbils (or specialized buds) :** Bulbils are modified and enlarged bud, meant for propagation. When bulbils detach from parent plant and fall on the ground, they germinate into new plants and serve as a means of vegetative propagation. *Example Agave* and *Allium proliferum*.

3.6.2 Types of Stem

Majority of angiosperm possess upright, vertically growing erect stem. They may be many types they are (i) Excurrent, (ii) Decurrent, (iii) Caudex and (iv) Culm.

i. Excurrent

The main axis shows continuous growth and the lateral branches gradually becoming shorter towards the apex which gives a conical appearance to the trees. Example: *Monoon longifolium(Polyalthia longifolia), Casuarina.*

ii. Decurrent

The growth of lateral branch is more vigorous than that of main axis. The tree has a rounded or spreading appearance. Example: *Mangifera indica.*

iii. Caudex

It is an unbranched, stout, cylindrical stem, marked with scars of fallen leaves. Example: *Cocos nucifera.*

iv. Culm

Erect stems with distinct nodes and usually hollow internodes clasped by leaf sheaths. Example: Majority of grasses including Bamboo.

3.6.3 Modification of Stem

I. Aerial modification of stem

1. Creepers

These are plants growing closer (horizontally) to the ground and produce roots at each node. Example: *Cynodon dactylon, Centella.*

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